SUMMER TERM 2021

ECON0039: ADVANCED MACROECONOMICS
Solution (April 7, 2023)

Coursework assessment

All questions should be answered.

Part A carries 20% of the total mark, part B 40% and part C 40%. Questions B.1 and B.2 have the same weight.

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PART A : Monetary Policy and Sticky Prices in a "UV" model

This problem considers a monetary model in the UV setting of Lecture 4 on Business Cycles. The economy features a representative firm, a representative household and a Central Bank. The representative household owns the firm and receives its profits π. There is a unique produced good that can be used for consumption c and investment x. Its price is p. Money is the numéraire good. It is supplied in quantity $\overline{m}$ by the Central Bank and is distributed to the housholds.

The representative household utility is of the UV form, and is given by

$$U\left(c, \ell, \frac{m}{p}\right) + V(x, \theta) = \ln c - B\ell + \nu \ln\left(\frac{m}{p}\right) + \theta \ln x,$$

where m is the money demand of the household, ℓ is labour. Agents demand money because $\frac{m}{p}$ brings some utility.

The household budget constraint is

$$pc + px + m \leq w\ell + \overline{m} + \pi$$

where w is the nominal wage.

The representative firm produces a single good according to the technology $y = A\ell^\alpha$ with $0 < \alpha < 1$. y is split into consumption c and investment x.

Finally, all markets are competitive.

Assume first that prices are flexible.

A.1 Write the profit maximization problem of the firm and the first order conditions to this maximization. Derive good supply y, labour demand ℓ^d and profit π as a function of w, p and parameters.

The representative firm solves the problem:

$$\max_{\ell} \pi = py - w\ell = pA\ell^\alpha - w\ell,$$

and first order condition is

$$\alpha pA\ell^{\alpha-1} - w = 0,$$

from which we obtain $\boxed{\ell^d = \left(\frac{1}{\alpha A}\frac{w}{p}\right)^{\frac{-1}{1-\alpha}}}$ and $\boxed{y = A^{\frac{1}{1-\alpha}}\left(\frac{1}{\alpha}\frac{w}{p}\right)^{\frac{-\alpha}{1-\alpha}}.}$ The expression for profit is then $\boxed{\pi = (1-\alpha)\alpha^{\frac{\alpha}{1-\alpha}}A^{\frac{1}{1-\alpha}}p^{\frac{1}{1-\alpha}}w^{\frac{-\alpha}{1-\alpha}}.}$

A.2 Write the utility maximization problem of the household and the four first order conditions to this maximization, with respect to c, x, ℓ and m. Derive consumption demand c, investment demand x, money demand m and labour supply ℓ^s as a function of w, p, π, $\overline{m}$ and parameters.

The representative household solves the problem:

$$\max_{c,\ell,m,x} \ln c - B\ell + \nu \ln\left(\frac{m}{p}\right) + \theta \ln x,$$

subject to the budget constraint

$$pc + px + m \leq w\ell + \overline{m} + \pi \qquad (\lambda).$$

First order conditions are

$$\frac{1}{c} = \lambda p \qquad (1)$$
$$B = \lambda w \qquad (2)$$
$$\frac{\nu}{m} = \lambda \qquad (3)$$
$$\frac{\theta}{x} = \lambda p \qquad (4)$$

(1) and (3) give $pc = \frac{m}{\nu}$. (1) and (4) imply $x = \theta c$. Putting all together, we obtain $\boxed{c = \frac{1}{B}\frac{w}{p}}$, $\boxed{m^d = \nu \frac{w}{B}}$ and $\boxed{x = \frac{\theta}{B}\frac{w}{p}}$. Total good demand is then $\boxed{c + x = \frac{1+\theta}{B}\frac{w}{p}}$. Plugging those three expressions in the budget constraint, we obtain $\boxed{\ell^s = \frac{(1+\theta+\nu)}{B} - \frac{(\overline{m}+\pi)}{w}}$.

A.3 Define the general equilibrium of this economy. Find the equilibrium levels of quantities c^E, ℓ^E, x^E, y^E and m^E, wage w^E and price p^E.

A general equilibrium is a collection of prices and quantities such that, *(i)* given wage w^E and price p^E, quantities c^E, ℓ^E, x^E, y^E and m^E maximise utility and profits, *(ii)* good, labour and money markets clear.
The table below gives the expressions for demand and supply on the three markets, as obtained above.

	Supply	Demand
Good market:	$y = A^{\frac{1}{1-\alpha}}\left(\frac{1}{\alpha}\frac{w}{p}\right)^{\frac{-\alpha}{1-\alpha}}$	$c + x = \frac{(1+\theta)}{B}\frac{w}{p}$
Labour market:	$\ell^s = \frac{(1+\theta+\nu)}{B} - \frac{(\overline{m}+\pi)}{w}$	$\ell^d = \left(\frac{1}{\alpha A}\frac{w}{p}\right)^{\frac{-1}{1-\alpha}}$
Money market:	$\overline{m}$	$m^d = \nu\frac{w}{B}$

where $\pi = (1-\alpha)\alpha^{\frac{\alpha}{1-\alpha}} A^{\frac{1}{1-\alpha}} p^{\frac{1}{1-\alpha}} w^{\frac{-\alpha}{1-\alpha}}$.

To solve for the equilibrium, $y = c+x$ implies $\boxed{\frac{w^E}{p^E} = \alpha^\alpha A\left(\frac{B}{1+\theta}\right)^{1-\alpha}}$. Then labour demand gives $\boxed{\ell^E = \alpha\frac{1+\theta}{B}}$. Using the demands c and x, one gets $\boxed{c^E = \alpha^\alpha A\frac{B^{-\alpha}}{(1+\theta)^{1-\alpha}}}$ and $\boxed{x^E = \theta\alpha^\alpha A\frac{B^{-\alpha}}{(1+\theta)^{1-\alpha}}}$. From the production function, the expression for ℓ^E implies $\boxed{y^E = \alpha^\alpha A\left(\frac{1+\theta}{B}\right)^\alpha}$. Using money demand with $m^d = \overline{m}$, we obtain $\boxed{p^E = \overline{m}\frac{(1+\theta)^{1-\alpha}B^\alpha}{\alpha^\alpha A\nu}}$. Finally, using the expressions for the equilibrium real wage and for the equilibrium price, we obtain $\boxed{w^E = \frac{B}{\nu}\overline{m}}$.
Note that money supply $\overline{m}$ does not affect real quantities, and increases linearly p^E and w^E. Money is neutral.

A.4 Can a technology shock ($\partial A > 0$), an investment demand shock ($\partial\theta > 0$) or a monetary policy shock ($\partial\overline{m} > 0$) create business cycles comovements? Discuss.

It is easy to check that shocks have the following effect:

	∂c^E	∂x^E	∂y^E	$\partial \ell^E$
$\partial A > 0$	+	+	+	0
$\partial \theta > 0$	-	+	+	+
$\partial \overline{m} > 0$	0	0	0	0

As we can see, money is neutral and monetary shocks do not generate business cycles. An investment demand shock does increase x, but not c, which is counterfactual as consumption and investment are procyclical in the data. A technology shock does increase all quantities. It leaves employment constant, but as discussed in class, it is possible to obtain an increase in employment if we change the utility function so that wealth and substitution effect do not exactly compensate (at the cost of less tractability). But in the data, technological shocks are not procyclical. Therefore, the flex price model does not propose a realistic theory of the business cycle.

Assume now that the price of the good is fixed at a level $p^F > p^E$. Let's consider a fix-price "equilibrium" in which the money market clears, but the good and labour market may not. We assume *voluntary exchange*, meaning that households are not forced to work or spend more than what is optimal for them, neither are firms forced to hire or produce more that what is optimal for them.

Because $p^F > p^E$, the structure of the equilibrium will be as follows. Money market will be in equilibrium, which will determine equilibrium wage w^F. Output y^F will be determined by good demand $c^F + x^F$. To that level of output will correspond a demand for labour ℓ^F. That level of labour demand will be the equilibrium employment.

A.5 Find the fix-price equilibrium values $c^F, \ell^F, x^F, y^F, m^F$ and w^F.

Demand and supplies have the same expression as before and p^F is given. Clearing the money market gives $\boxed{m^F = \overline{m}}$ and $\boxed{w^F = \frac{B}{\nu}\overline{m} = w^E.}$ The real wage is therefore $\frac{w^F}{p^F} = \frac{w^E}{p^F} < \frac{w^E}{p^E}$ because $p^F > p^E$. Output is determined by good demand, so that $y^F = c^F + x^F$. Using the demands for c and x, we obtain $\boxed{c^F = \frac{1}{\nu}\frac{\overline{m}}{p^F},}$ $\boxed{x^F = \frac{\theta}{\nu}\frac{\overline{m}}{p^F}}$ and $\boxed{y^F = \frac{1+\theta}{\nu}\frac{\overline{m}}{p^F}.}$ Employment is then obtained using the production function: $\boxed{\ell^F = \left(\frac{1+\theta}{A\nu}\frac{\overline{m}}{p^F}\right)^{\frac{1}{\alpha}}.}$

A.6 Can a technology shock ($\partial A > 0$), an investment demand shock ($\partial \theta > 0$) or a monetary policy shock ($\partial \overline{m} > 0$) create business cycles comovements in that fix-price economy? Discuss.

It is easy to check that shocks have the following effect:

	∂c^F	∂x^F	∂y^F	$\partial \ell^F$
$\partial A > 0$	0	0	0	-
$\partial \theta > 0$	0	+	+	+
$\partial \overline{m} > 0$	+	+	+	+

Clearly, a technological shock cannot create business cycle movements, as output and employment are demand determined. The investment demand shock θ could be a candidate as it increase all variables without strictly decreasing any. The monetary shock, as in IS-LM, is here expansionary and create business cycles like movements.

PART B : Questions

ARE the following statements true, false or uncertain? You should give a structured answer in no more than one page.

B.1 Medical improvements increase income per capita and population in the Malthusian regime.

FALSE

- What is the Malthusian Regime? It corresponds to the functioning of human societies before the industrial revolution. In short, income per capita is roughly constant in the Malthusian regime, close to subsistence level. Changes in the availability of resources translate into population changes.
- In the Malthusian regime, average growth rate of income per capita is negligible : 0.05% a year between 1000 and 1820 (Maddison).
- Increases in resource translate into increases in population. World population was 231 millions in 1 AD and 268 millions in 1000 AD, then 438 millions in 1500 AD. Resource expansions after 1500 implied an increase in population growth (1041 millions people in 1820)
- An example of the functioning of the Malthusian regime is given by the English Black Death (1348-1349) and its aftermath. Following the English Black Death, English population went down from 6 millions to 3.5 millions. An increase in the land/labor ratio followed, so that the real wage tripled for 150 years. This caused an increase in population, so that in the 1560's, the real wage was back to the pre-plague level.
- An increase in medical knowledge is likely to increase population in the long run. But as resources are scarce, this permanent increase in population will lead to a permanent *decrease* in income per capita.
- One can formally illustrate this mechanism with the CLARK's model we have seen in class. The model has three equations:

Demography (1): the birth rate b is high when income per capita y is high.

$$b(t) = b\underbrace{(y(t))}_{+}$$

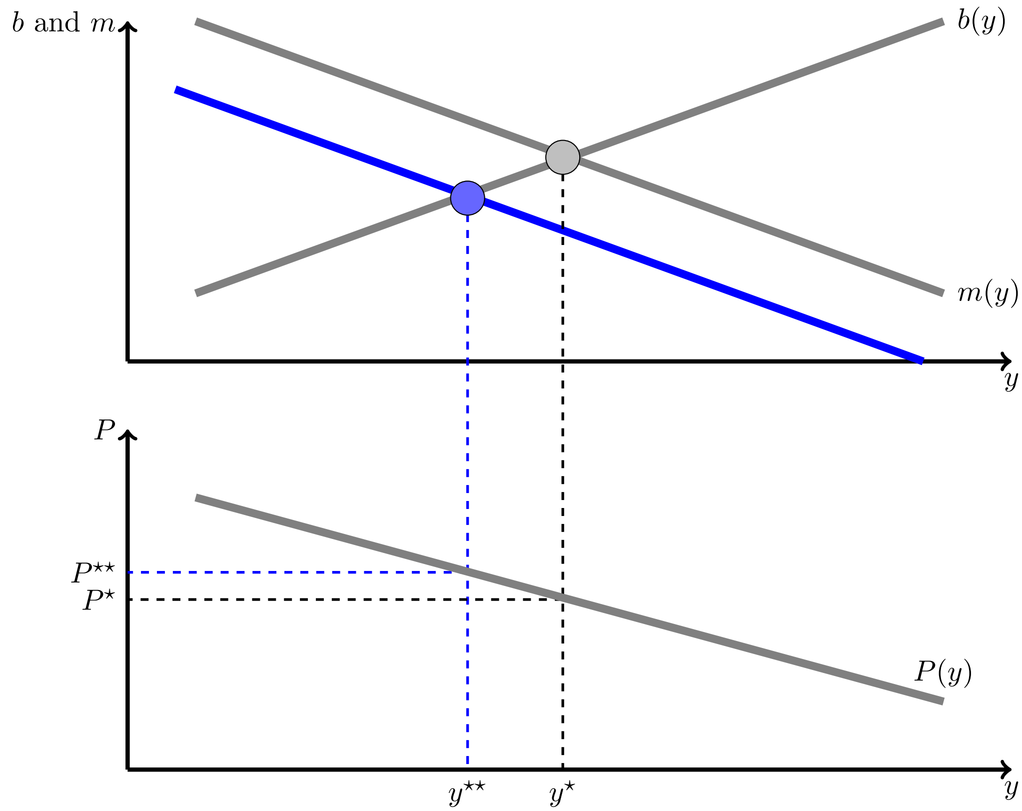
Demography (2): the mortality rate m is low when income per capita y is high.

$$m(t) = m\underbrace{(y(t))}_{-}$$

Technology: Population P is low when income per capita y is high.

$$P(t) = P\underbrace{(y(t))}_{-}$$

– Graphically, medical progress will shift the mortality locus downward (the new blue line), which will decrease income per capita to $y^{\star\star}$ and increase population at the new steady state $P^{\star\star}$, as shown on the figure.



B.2 High inflation reduces money demand *ceteris paribus*.

TRUE

- One can present money demand either by referring to the analysis of Keynes in the General Theory, or by refeering to the Krugman's model with a Cash-in-Advance constraint, or more generally by describing what is the use and cost of money. Here I am taking the third alternative.
- Let's restrict to households money demand (firms also use money, for very similar motives). Money is an asset that is used as a mean of payments and as a store of value. The main property of money is that it is infinitely liquid, meaning that it can be used instantaneously and at no costs to make payments, which is not the case for example of saving accounts (that need time to be liquidated) or houses (that need time plus transaction costs to be sold), and increasingly so with their income, in order to make transactions.
- On the other side, money has an *opportunity cost*. The more money is held, the less is held using assets that bring an interest. Take bonds as the alternative asset. Bonds pay the nominal interest rate i. By holding money instead of bonds, households give up i per period and per pounds held. The opportunity cost of money is therefore the nominal interest rate.
- By arbitrage between all existing assets, the nominal interest rate i is equal to the real interest rate r plus expected inflation π^e. This is known as the Fisher relation: $i = r + \pi^e$.
- Therefore, an increase in inflation (more precisely *expected* inflation) will increase the opportunity cost of money, and threrefore decrease money demand *ceteris paribus*.

PART C : Discussion

Below is an article published in The Economist in November 2020 and entitled "Deconstructing the Lebanese central bank's Ponzi scheme" As always in The Economist, the article is not signed. Read carefully this article before answering the questions below. For each question, you should propose a structured answer.

C.1 Write a 10-line summary.

- The August 4th 2020 explosion in the Port of Beirut is only the most spectacular sign of the deep crisis that Lebanon has been going through recently, with annual inflation at 120% in August 2020, the Lebanese pound loosing much of its value against the dollar (9000 to the dollar against 1500 a few years back) and GDP shrinking by a quarter in 2020.
- After the civil war, the peg to the dollar was the condition for confidence and recovery.

- That peg was not defended with oil revenues (Lebanon has none), but using the vibrant Lebanese financial sector, through a quite creative "financial engineering"; a system in which the Banque du Liban financed trade and public deficit by serving high interest rates on dollar denominated deposits.
- That system cannot be sustained in the long run, as it is a typical Ponzi scheme.
- Reforms are unfortunately not on their way, partly because of the strong ties between the Lebanese politicians and bankers.

C.2 Use the three figures of that article to assess Lebanon macro-financial situation. (no longer than one page (roughly 500 words))

- As a first comment, let's note that it would have been better if those figures had been in constant Lebanese pounds, or in percentage of nominal GDP.
- Top left panel shows a large increase in nominal deposits at commercial banks along the 2010's, then massive withdrawals of funds from 2019 and onwards. The drop is about 20% since 2019. Because they were running out of cash, commercial banks had to limit dollar withdrawals. For example, credit cards now face a spending limit of $50.
- Top right panel shows commercial banks deposits at the Banque du Liban (BdL). We also observe a large increase in the 2010's. This happened because the BdL was serving high (excessive) interest rates (the spread was 11%) on those deposits. Similarly, there is an abrupt reduction of theses deposits in 2019, although these withdrawals stabilised after.
- Bottom panel shows one of the cause of the current situation, namely Government indebtedness. We see on that graph an increase in the stock of Government bonds. The share of those bonds held by foreign investors declined, the share of those held by domestic investors and commercial banks stayed constant, while more and more of that debt was monetised –i.e. held by the central bank.
- The doom loop is then in place: monetising public debt lead to inflation; households tried to get rid of their Lebanese pounds and massively withdraw their deposits from commercial banks. As the pound depreciated against the dollar, banks faced massive losses. Then the question is who shall cover those losses, the depositors (mainly households) or the bank shareholders and the wealthy depositors? Because of the strong ties between politicians and bankers (for example, one of the biggest bank of the country, BankMed, belongs to a large extent to the family of the new prime minister Saad Hariri) and the banks lobbying, it seems that the former solution is being preferred to the latter.

C.3 Why is the current Lebanese situation typical of an episode of hyperinflation? (not more than roughly 500 words)

We find in the Lebanese situation all the ingredients of an hyperinflation episode:

- As we saw in the Germany of the 1920's and in Zimbabwe in the late 2000's, hyperinflation is most of the time triggered by large monetisation of public deficits. In the Weimar Republic and in Zimbabwe, the Government ran large deficits, did not collect much taxes, and financed its spending by issuing bonds that were subscribed by the central bank. This increased prices, leading households and firms to try to get rid of their stock of domestic currency and to hold foreign currencies (say US dollars). Therefore, the local currency depreciated against the dollar and the price level skyrocketed.
- The current situation is not different in Lebanon. The BdL has been financing most of the local-currency debt and of fiscal deficit.
- Furthermore, to maintain the peg *vis-à-vis* the dollar, the BdL has been attracting dollars by serving very high interest rates on dollar deposits (7% in 2019).
- Facing massive withdrawals, the banking system is now limiting access to dollars, at a depreciated rate (now 3,900 pounds to the dollar as compared to 1,500 previously). But on the black market exchange rate is not 3,900 but rather 9,000 pounds to the dollar.
- We also observe a political crisis. The former prime minister has resigned, the new one cannot form a government. Public assets seem to have lost much of their value. The years of bad public management had been spectacularly exemplified by the 2020 Beirut port explosion.
- To summarize, we are in the typical hyperinflation episode. As we saw in the lecture,
  - Hyperinflation happens in situation of political instability
  - Hyperinflation is always related with fiscal deficits that are financed by money creation (that are *monetized*)
  - Economic agents run away from money because they expect prices to increase even further, and therefore reduce the real value of money balances.
- The Cagan model of hyperinflation we have seen in class (and the rational expectations version of Sargent) show well that exploding money supply creates hyperinflation. The Cagan model, becasue expectations are adaptive, also shows the possibility of momentum in inflation, so that one could observe hyperinflation even though there is no more large increase of money supply.

C.4 Assume that you are leading an IMF mission to Lebanon. Which actions would you suggest to end hyperinflation. Why? What else would you suggest to do? (not more than roughly 500 words)

- Consider first the objective of ending hyperinflation.

- We have studied in class the German and Zimbabwean episodes and solved the CAGAN's model with adaptive expectations, but also the SARGENT's version with rational expectations.
- A few conclusions arise as how to end an hyperinflation episode.
- Fiscal discipline and/or an independent central bank are needed to prevent monetised deficits.
- One needs to anchor individuals' inflation expectations , in particular when a momentum-driven inflation break-out is possible.
- SARGENT pointed out four episodes (Austria, Hungary, Germany and Poland) in the 1920's where hyperinflation showed no momentum once policy was changed.
- Typically, commitment to not monetising the debt and "dollarization" are necessary conditions for rapidly ending up hyperinflation
- Part of the trick to anchor inflation expectations is for Government policy to be credibly seen as inflation fighters.
- A typical reform that Lebanon could adopt would be *(i)* to end capital controls (so that the economy can effectively become dollarized), *(ii)* to make the central bank independent so that it will stop monetising public debt, *(iii)* if needed change the currency to a "new pound" that would be worth, say, 10,000 old pounds (10,000 is roughly one dollar on the black market) and *(iv)* restructure the debt (increase maturity + haircut).

– Although these recommendations might end hyperinflation, they will for sure not solve Lebanese economic and financial crisis. On would most certainly want the IMF to care not only about inflation, but also about financial stability and growth.

– The anti-inflation policy will most likely lead to a collapse of the Lebanese banking system. And banks are needed for the economy to recover. The question, which is highly political, is who will take the hit? It could be the banks shareholders if banks go to bankruptcy or it could be the depositors if one decides to wipe out their deposits. One middle ground could be to guarantee, say, the first x pounds of deposits, and expropriate banks shareholders –i.e. to nationalize banks.

– Of course, this is about the short run. Long run growth needs sound institutions. Reforming Lebanese institutions is (*this is a personal opinion*) beyond what IMF can do, and shall be in the hands of Lebanese people.

"Deconstructing the Lebanese central bank's Ponzi scheme "
The Economist, Nov 5th 2020

A web of mutual liabilities binds the central bank to the government and the banks

George Azzi no longer allows customers inside his pharmacy. Too many shops have been robbed. Instead he takes orders through a plexiglass window. But now, he jokes, there is not much left to steal. A currency crisis has left his shelves half-empty. A nearby petrol station has rationed supplies, limiting drivers to 20 litres. The supermarket around the corner has not stocked fresh chicken for weeks because poultry farmers will not sell at the government-mandated price.

Lebanon has spent the past year lurching from one crisis to the next: protests, covid-19, the catastrophic explosion at Beirut's port on August 4th. In the background all along has been a grinding economic collapse. Pegged for decades at 1,500 to the dollar, the Lebanese pound recently traded on the black market as low as 9,000. Annual inflation hit 120% in August. GDP may shrink by a quarter this year.

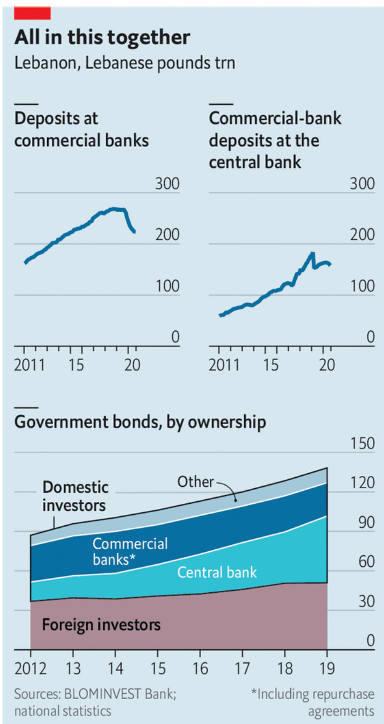
Subsidies have kept a few essentials affordable: medicine, fuel, a morning manoucheh (flatbread). But the central bank, the Banque du Liban (BdL), has less than $2bn in usable foreign-currency reserves remaining. By the year's end it will be unable to maintain its subsidy scheme, through which it grants importers of essential goods access to dollars at preferential rates. Some Lebanese have begun hoarding food and medicine in anticipation of far higher prices.

It will be the end of a monetary regime that once seemed inviolable. The peg helped restore confidence after a 15-year civil war, during which inflation peaked at 487% and the pound lost 80% of its value. But it was built on an unproductive economy and a complex series of transactions that, in recent years, came to resemble a Ponzi scheme. It has now collapsed. Yet Lebanon's political and financial elites–sometimes one and the same–are unwilling to bear the costs of fixing it. Instead they have forced the economy into a brutal self-correction, with the heaviest consequences piled on ordinary citizens.

Most Arab states with fixed currencies defend their pegs with revenue from oil and gas exports. Lebanon has none. It has few goods exports at all: at their peak in 2012 they came to $4.4bn, against $21.1bn in imports. Other sources of hard currency, such as tourism and property, were insufficient to sustain a peg, with current-account deficits exceeding 25%, and fiscal deficits of over 10%, of GDP.

It did have one reliable source of dollars: a vibrant financial sector that, at its peak in 2018, held 269trn pounds of deposits, then worth $179bn–more than three times the country's GDP. About two-thirds of deposits were in dollars. To funnel some of that into its coffers the BdL crafted a scheme it called "financial engineering". Starting in 2016 it exchanged local-currency debt with the finance ministry for dollar-denominated bonds, which it sold to commercial banks with high rates of return.

Banks could also borrow pounds against their dollar deposits at the BdL, then lend them back to it, with a spread of 11 percentage points on the transaction.



The longtime central-bank governor, Riad Salamé, defends the scheme as a short-term fix. In his telling the BdL did not create Lebanon's deficits: those were the work of politicians who failed even to pass a budget between 2005 and 2017. True enough–but the scheme made those debts possible and tied everyone in a web of mutual liabilities. By 2018 the BdL was financing most of the fiscal deficit, and by 2019 most of Lebanon's public debt (more than 80% of local-currency debt and 60% of foreign-currency bonds) was owned by banks and the BdL. The banks, in turn, deposited more than half their liquidity with the BdL. In its annual assessment of the Lebanese economy, the IMF reckoned that banks' combined exposure to the BdL and the government amounted to 69% of their total assets in May 2019, more than eight times their Tier 1 capital, largely due to the rise in their deposits with the central bank.

Sustaining this house of cards required an ever larger supply of fresh dollars, which necessitated ever higher yields to lure depositors. Average interest rates on dollar deposits climbed from 3% in 2016 to almost 7% three years later. But after a decade of growth, bank deposits flattened off in late 2018, in part because low oil prices hurt the Gulf economies where many Lebanese work. Last year they began to decline. Short of cash, banks severely curtailed dollar withdrawals. Credit cards that once financed trips to Europe now carry spending limits of $50. In March the state defaulted when it failed to make a payment of $1.2bn on a hard-currency bond. The prime minister at the time, Hassan Diab, estimated an $83bn loss in the banking sector.

Most customers can withdraw their dollar savings only in local currency (Dan Azzi, a former executive at Standard Chartered, a bank, dubbed these "lollars"). Since April the BdL has allowed withdrawals at an above-market exchange rate, currently 3,900 pounds to the dollar. That is still up to 55% below the black-market rate–"a massive haircut on the Lebanese," says Alain Bifani, who resigned in June after 20 years as the director-general of the finance ministry. Despite the haircut, people have rushed to withdraw deposits. Cash in circulation was up 274% in the year to August.

Mr Diab proposed shifting more of the losses on to banks, rather than entirely on their depositors. His cabinet approved a plan in April calling for a bail-in, wiping out shareholders and imposing a haircut on wealthy depositors. He also began talks with the IMF about a financial agreement worth up to $10bn. The fund was broadly supportive of his plan, but it was doomed by lobbying from banks. They have strong allies in parliament: most of Lebanon's big banks are at least partly owned by politicians or their families. In meetings with the IMF, bankers would not even agree on the scale of Lebanon's losses, insisting that Mr Diab's numbers were exaggerated.

Mr Diab has since resigned as prime minister, though he remains caretaker. His designated successor, Saad Hariri, who is still struggling to cobble together a government, is unlikely to take up his ideas. He had been prime minister before, when the financial-engineering scheme was in full swing, and his family has a stake in BankMed, one of the country's largest banks (his stepmother serves as vice-chair). Instead the association of banks has suggested selling $40bn in state assets to help clean up the mess. Critics argue this wildly overstates the value of Lebanon's assets, among them an electric company that cannot provide 24-hour electricity and a telecoms monopoly that offers some of the world's slowest internet speeds. Neither is likely to prompt a bidding war when a bankrupt state dumps them in a fire sale.

The BdL, once again, is left to pursue its policies, some of which seem to be attempts to keep the Ponzi scheme going. In August it told banks to boost their capital by 20% or "exit the market" in February next year. Depositors who have transferred at least $500,000 abroad since 2017 would be encouraged to repatriate 15% of that sum, and 30% for politically exposed persons. Banks may offer "incentives" to grease the wheels. What would incentivise customers to send money to an insolvent banking system was left unclear. Efforts to audit the central bank, meanwhile, have met with months of obstruction.

Even the good news is bad. The trade deficit fell by 51% in the first quarter, but mainly because few Lebanese can afford imported goods. Banks have fewer liabilities–customer deposits were down by 16% in August, compared with the previous year–but partly because depositors who need cash are taking haircuts to withdraw devalued pounds. Politicians and bankers still hope to avoid a reckoning, even as the economy crumbles around them. But they have nowhere left to turn.